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Beyond EHRs: External Clinical knowledge and cohort Features for medication recommendation

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ABSTRACT

Medication recommendation plays an important role in healthcare by supporting clinical decision-making, while rich clinical experiences and knowledge are key factors that contribute to the success of this task. However, existing methods still face certain limitations: the acquisition of clinical experience is confined to the historical records of an individual patient, and encoded clinical knowledge is typically used as coarse auxiliary information to enhance encoded records. To address these limitations, we propose the **EX**ternal **C**linical knowl**E**dge and coho**R**t **F**eatures (EXCERF) model for the recommendation task. EXCERF constructs an external Memory Neural Network shared across all patients to capture cohort features tailored to specific patient groups, allowing patients to access relevant clinical experiences distributed across a large cohort. Then the model incorporates clinical knowledge tuples as additional biases to refine interactions between clinically related entities within the self-attention mechanism, concurrently considering detailed clinical and semantic knowledge to build patient representations. Finally, EXCERF integrates representations of multiple admissions with GRU to generate the final recommendation. Experimental results on real-world clinical records demonstrates that EXCERF achieves superior performance and facilitates effective medication recommendation.

1. Introduction

The widespread adoption of Electronic Healthcare Records (EHRs) represents a major breakthrough in healthcare research, while recent technological innovations in deep learning have added significant impetus to various healthcare applications [1–4]. EHRs hold great promise for enhancing diagnostic and therapeutic outcomes due to the ability to monitor fluctuations in patient conditions in terms of multiple measures, and the wealth of patient data provides physicians with a vast amount of information to gain deep insight into patients, thereby supporting informed decision-making about care. Meanwhile, researchers have proposed a wide range of deep learning approaches to build healthcare applications, and with the potential to mitigate risks posed by the fast-paced nature of hospital work, medication recommendation [5–7] has emerged as a critical area of focus. Medication recommendation generates prescriptions as direct decision-making references to assist doctors in handling rapid changes in patient conditions, ultimately improving the efficiency of diagnosis and treatment. This has propelled medication recommendation to the forefront of clinical research, with significant implications for patient care and promising prospects for future applications.

As one of the primary perspectives contributing to the design of healthcare applications, predictive analytics discerns hidden patterns contained in EHRs [8] to build informative patient representations customized to their health conditions. While these approaches emphasize the meticulous encoding of individual patient EHRs to capture underlying patterns, they overlook two critical resources that doctors rely on in real-world clinical settings:

Clinical Experience: Experienced doctors highly regard EHRs of patients across various categories as crucial references, considering inherent collective features within these EHRs as important clinical experiences. These cohort features assist doctors in assessing potential changes in patient conditions to guide prescribing, thereby playing a pivotal role in medication recommendation by augmenting patient representations. While existing methods reference encoded EHRs from patients' own historical visits to enhance their representations [7,9,10], they often overlook cohort features and exhibit limited ability to integrate clinical experiences spanning diverse patients.

Clinical Knowledge: Clinical knowledge, such as disease relationships presented by International Classification of Diseases (ICD) system,

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enables doctors to consider the intricate interactions between various entities when analyzing EHRs of patients for diagnosis. Therefore, it is essential to incorporate clinical knowledge into the construction of entity representations to facilitate information exchanges. Meanwhile, given the structural similarity between sequences of medical entities and sentences in natural language, the prevalent longitudinal-based EHRs encoding strategy primarily draws inspiration from Natural Language Processing (NLP) methods to encode EHRs. These approaches rely primarily on semantic correlations rather than clinical knowledge to facilitate interactions between them [11–13].

Therefore, it is essential to integrate clinical experience and clinical knowledge into EHRs encoding to construct informative patient representations for medication recommendation. However, certain challenges still persist. Incorporating insights from other patients requires meticulous consideration of inter-patient correlations to identify relevant patient cohorts, while the inherent complexity of latent patient conditions [14] and exponential growth in computational demands as patients increase further complicate the task to extract cohort features. Furthermore, leveraging clinical knowledge to facilitate entity interactions requires nuanced alignment and integration of clinical knowledge with semantic correlation among medical entities, but the disparity in the form of the two types of information, graphs and sequences, often leads previous methods to encode knowledge separately and provide the results as initial inputs or supplementary queryable features [10, 15].

In light of the aforementioned limitations, we propose a novel medication recommendation model called **EX**ternal **C**linical knowl**E**dge and coho**R**t **F**eatures (EXCERF), aimed at enhancing personalized patient representations with knowledge-based interaction and clinical experience for effective prescribing. First, EXCERF transforms clinical reachability among entities indicated by ICD systems into diverse knowledge tuples, then the model integrates corresponding tuples into self-attention weight calculation within the Transformer framework, allowing entities to consider influences of clinically associated ones to strength connections of their representations. Second, EXCERF constructs an external key–value Memory Neural Networks (MemNN) shared across all patients, whose keys and values are implemented with linear layers. This allows the model to capture and leverage shared cohort features associated with patients of same categories, yet distributed across different individuals, without subjecting itself to significant computational burdens or complicated definition of patient correlations. Finally, EXCERF encodes visit representations enhanced by clinical knowledge and experience via Gated Recurrent Units (GRU) to generate prescriptions as final outcomes.

The contributions of EXCERF could be summarized as follows:

- EXCERF implements the external MemNN shared across all patients with linear layers to discern cohort features as clinical experiences, providing patients with informative attributes across diverse patient categories through the attention mechanism without incurring exponential computational costs.
- EXCERF identifies clinical relationships among medical entities within clinical code systems, and seamlessly integrates the knowledge presented as tuples into the self-attention mechanism, enabling TOKEPER to simultaneously consider clinical knowledge and semantic correlations to enhance patient representations.

The rest of the paper is structured as follows: Section 2 provides a review of related work. Section 3 describes all details of EXCERF. Section 4 presents a comprehensive discussion and analysis of experiments. Section 5 concludes the work and outlines potential directions for future research.

2. Related work

2.1. Transformer for representation update

Transformer [16,17] has achieved state-of-the-art performance in a wide range of tasks, and the self-attention mechanism that enables the model to capture contextual information from the entire input sequence plays an important role in the success. Nevertheless, calculating the attention weight between any tokens arbitrarily would make some of them attend to those ones that are not closely related to them, which would lead to distraction of attention weights and introduce noises during the information integration. Recently, different variants have been proposed to mitigate the limitation, such as local attention, stride attention and global attention [18], which select attended tokens with local windows or strides instead of building the complete attention matrix. These strategies select neighboring tokens empirically and lack guarantees for high attention weights, and they might still pick irrelevant tokens. Some other approaches then choose to rely on clustering of attention instead [19,20]. Furthermore, there are also methods that train additional networks to generate latent permutations over sequences [21], so that they can compute quasi-global attention with local windows.

These advances illustrate the capability of Transformer to model complicated dependencies via self-attention. Meanwhile, the integration of clinical knowledge is crucial for enabling information exchange between clinically relevant entities. However, existing methods fail to explicitly incorporate the knowledge into self-attention mechanism to construct informative patient representations, leading to inadequate encoding of EHRs.

2.2. MemNN-based feature extraction

MemNN consists of extra memory components that can be read and written to, which is used to extend existing neural network models by storing informative facts from extra knowledge bases or historical data [22]. But these components are not easy to train via backpropagation, so an end-to-end variant of MemNN is proposed to make the model more generally applicable in realistic settings [23]. Furthermore, conducting multi-hop reading on MemNN is a common choice to extract contextual information [24], since new evidence will be combined with the query to retrieve more informative knowledge in the subsequent reading process. In addition, MemNN is extended into the form of keys and values [25], so that it can encode extra knowledge with greater flexibility via nontrivial transforms between keys and values. MemNN has also been applied on medication recommendation. Previous researches used the strategy to model intra-view and inter-view interactions in EHRs [26], contextual information in interactions between patients and doctors [27], and global as well as personalized features of patients [28].

MemNN serves as a critical tool to encode auxiliary information that acts as important source of knowledge to enhance encoding results. However, challenges remain in capturing cohort features that distributes across patients and are not explicitly available, which requires efficient strategies to obtain inherent features within different patient categories without incurring significant computational overhead.

2.3. Medication recommendation

Medication recommendation attracts significant attention as an important healthcare application, and drives various advances in deep learning methodologies. Longitudinal-based methods use RNN to extract temporal patterns in EHRs to build patient representations for the recommendation [10]. Moreover, MemNN is applied to store encoded EHRs from previous visits, enabling those models to reference clinically related visits to improve performance [11,12]. External knowledge beyond EHRs also plays a crucial role in enriching patient representations.

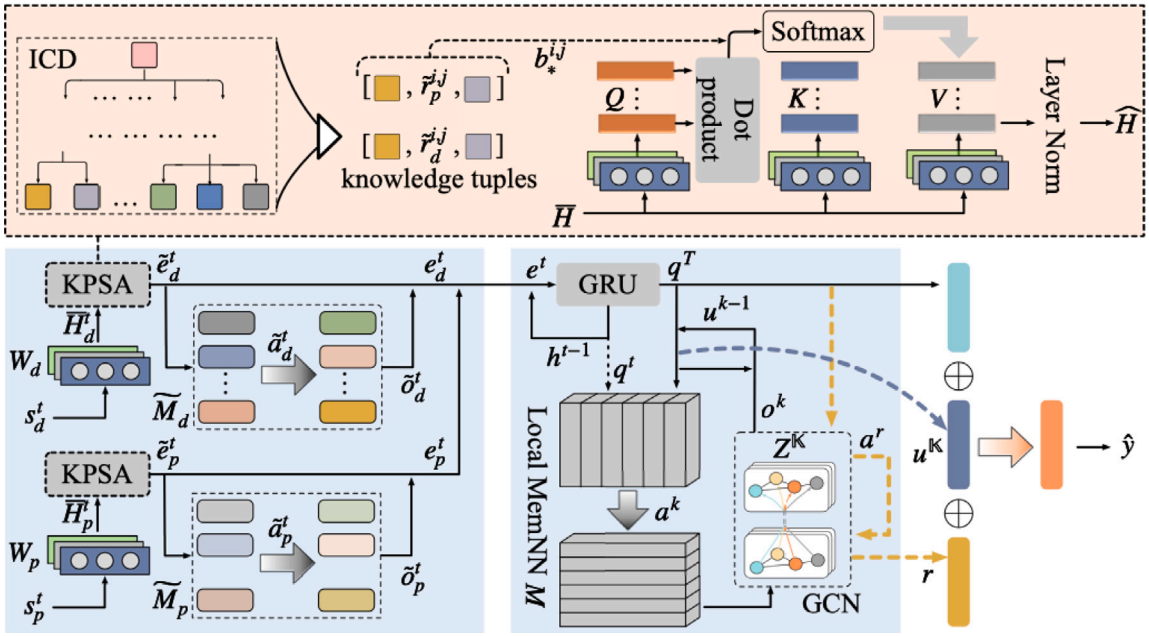


Fig. 1. The workflow of EXCERF. The model applies the Knowledge-Powered Self-Attention (KPSA) module to integrate clinical knowledge with EHRs, and the external MemNN learns cohort features spread across patients to enhance admission representations as clinical experience. The enhanced representations are subsequently encoded via GRU, while a local MemNN is built for each patient to make use of their historical prescriptions.

Typically encoded using GNN, such knowledge forms initial inputs or queryable memory banks to provide pertinent information [11,15,29], while hierarchical dependencies among entities within knowledge are also exploited to build effective reasoning paths when prescribing [7,30]. Furthermore, Reinforcement Learning (RL) provides an alternative approach to medication recommendation [31]. It can incorporate clinical knowledge into the reward calculation to avoid drug interactions [32], while the off-policy actor–critic RL framework applies the actor network refined by indicator and evaluation signals to recommend treatments, with the critic network evaluating actor's decisions to guide the learning process [33]. Additionally, treating medication prediction as an order-free Markov Decision Process, RL can integrate patient data via Dual-CNN to learn correlative interactions between medications for effective recommendations [34].

Despite these advancements on medication recommendation, prior studies often neglect cohort features distributed across patients, which serve as clinical experience in real-world clinical setting. Moreover, clinical knowledge is encoded and applied as initial inputs or queryable auxiliary information, rather than facilitating clinically meaningful interactions between entities within EHRs during encoding.

These gaps motivate this study to: (1) develop efficient approach to capture and apply cohort features without imposing excessive computational demands; (2) integrate clinical knowledge to explicitly enhance connections between relevant entities for clinically meaningful interactions. By addressing these issues, the proposed EXCERF aims to conduct effective medication recommendation by constructing informative patient representations.

3. Proposed method

In its pursuit of effective medication recommendation, EXCERF begins by exploring clinical knowledge presented as tuples, and then integrates these tuples into the self-attention mechanism to facilitate information communication among those entities. After that, EXCERF builds an external MemNN accessible to the entire patient cohort, providing cohort features as clinical experience to enhance patient representations. Finally, those representations are encoded using GRU for prescription generation, while a local MemNN is tailed for each patient to serve as a reference point for historical prescriptions. The overview is demonstrated in Fig. 1.

3.1. Notations

The clinical records of an individual patient consist of three categories of distinctive observations, comprising diagnostic and procedures denoted as ICD codes, as well as medications presented in the form of Anatomical Therapeutic Chemical (ATC) code. These codes are subsequently converted into three fixed vocabularies S_d, S_p and S_m, each assigning a unique integer to represent its corresponding entity. Thus, the records of a patient $X = [x^1, x^2, \ldots, x^T]$ can be explained as a sequence of multivariate observations, where T is the number of admissions, and x^t represents the records of the tth admission, encompassing diagnostic $d^t \subseteq S_d$, procedures $p^t \subseteq S_p$ as well as medications $m^t \subseteq S_m$. When utilized as input, EXCERF further transforms these records into multi-hot vectors s_d^t, s_p^t and s_m^t. For simplicity, we use S_* and s_*^t to indicate the unified definition for different types of medical entities. We also adopt this symbol $*$ in the subsequent introduction of different types of medical entity notations to simplify the description.

The ICD system serves as a fundamental tool in the categorization of diseases and procedures, offering a hierarchical structure O_* to classify various medical entities, including diagnoses and procedures, into different categories. Within this structure, these entities are represented as leaf nodes, and the shortest path connecting two such nodes inherently reveals the clinically meaningful relations between these entities. Therefore, given the vocabulary S_* and ontology O_*, such relation can be manifested as a knowledge tuple $\tilde{k}_* = (\tilde{h}_*, \tilde{r}_*, \tilde{t}_*)$, where $\tilde{h}_* \in S_*$ and $\tilde{t}_* \in S_*$ are the two medical entities, and $\tilde{r}_*$ is associated relation weight calculated based on O_*.

Following the incorporation of knowledge tuples into the self-attention mechanism to encode s_*^t, EXCERF derives the result $\hat{H}_*^t = [\hat{h}_*^1, \hat{h}_*^2, \ldots, \hat{h}_*^n]$. Subsequently, the model integrates $\hat{H}_*^t$ into a comprehensive representation $\tilde{e}_*^t$, which would then interact with the external key–value MemNN $\widetilde{M}_*$ to read cohort features $\tilde{o}_*^t$. The model first calculates the attention between $\tilde{e}_*^t$ and the learnable key $\widetilde{M}_{*,key}$, and then read relevant cohort features from the value $\widetilde{M}_{*,key}$ based on the attention. Finally, the model combines $\tilde{e}_*^t$ and $\tilde{o}_*^t$ to obtain the record representation e_*^t.

Given the representation $e_*^t (t \in [1, T])$ that enhanced with clinical knowledge and experience of all T admissions of a patient, EXCERF encodes them with GRU to capture temporal dependencies across multiple

Table 1
Notations used in EXCERF.

Notations	Description
S_*	Vocabularies of type $*$
X	Patient records
x^t	Patient records of the tth admission
d^t	Diagnoses of the tth admission
p^t	Procedures of tth admission
m^t	Medications of tth admission
s^t_*	Multi-hot vector of type $*$
T	Number of patient admissions
O_*	Hierarchical structures of type $*$ in ICD
$\tilde{k}_*$	Knowledge tuple of type $*$
$\tilde{K}_*$	Knowledge tuple set of type $*$
$\hat{H}^t_*$	Representations of type $*$ for the tth admission considering knowledge tuples
$\hat{h}^n_*$	Representation of the nth entity in $\hat{H}^t_*$
$\tilde{e}^t_*$	Comprehensive representation of $\hat{H}^t_*$
$\widetilde{M}_*$	External MemNN of type $*$
$\widetilde{M}_{*,key}$	Keys in $\widetilde{M}_*$
$\widetilde{M}_{*,value}$	Values in $\widetilde{M}_*$
$\tilde{o}^t_*$	Cohort features read from $\widetilde{M}_*$ according to $\tilde{e}^t_*$
e^t_*	Representation of type $*$ for the tth admission
q^t	Representation of the tth admission
M	Local MemNN
o^k	Feature read from M at the kth hop of reading
$\hat{Y}$	Prescription generated by EXCERF

admissions, yielding the updated representation $q^t (t \in [1, T])$. Then the model uses q^t as well as $m^t (t \in [1, T-1])$ of the $T-1$ historical admissions as keys and values respectively to construct the distinct local MemNN M tailored to each patient. When generating recommendations for the Tth admission, this design allows EXCERF to use q^T as the query to retrieve encoded prescriptions o^k related to conditions similar to q^T. Subsequently, o^k is merged with q^T to generate the final prescription. Therefore, given the historical records $[x^1, x^2, \ldots, x^{T-1}]$ of the $T-1$ historical admissions, diagnoses d^T and procedures p^T of the current Tth admission, knowledge tuple set $\tilde{K}_*$, the external MemNN $\widetilde{M}_*$, EXCERF aims to recommend prescription $\hat{Y}$.

These notations are summarized in Table 1.

3.2. Medical entity embedding

Following the conversion of medical codes into fixed vocabularies S_d and S_p, EXCERF transforms diagnoses d^t and procedures p^t into the corresponding multi-hot vectors s^t_d and s^t_p respectively. Then the model encodes these multi-hot vectors to obtain continuous representations $\overline{H}^t_d$ and $\overline{H}^t_p$ as follows:

$$\overline{H}^t_d = W_d s^t_d \tag{1}$$

$$\overline{H}^t_p = W_p s^t_p \tag{2}$$

where $W_d \in \mathbb{R}^{|S_d| \times dim}$ and $W_p \in \mathbb{R}^{|S_p| \times dim}$ are the embedding matrices, while dim denotes the dimension of representations. These embedded representations then act as the initial inputs to the model for subsequent processing.

3.3. Knowledge-powered self-attention

EXCERF leverages Knowledge-Powered Self-Attention (KPSA) to encode records of each individual admission, which simultaneously takes contextual correlations and the interaction according to clinical knowledge between entities into account. KPSA begins by exploring clinical knowledge presented by the ICD system. The system employs a hierarchical structure to classify diseases and procedures into different categories based on their clinical relations, and those categories at each level contain diseases or procedures with varying degree of correlations. The final classification result can be seen as a tree structure O_*, which enables ICD to assign unique alphanumeric codes to those diseases and procedures to demonstrate their categories and correlations. Therefore, according to the shortest path between the corresponding leaf nodes of two entities, a knowledge tuple $\tilde{k}_* = (\tilde{h}_*, \tilde{r}_*, \tilde{t}_*)$ can be built to present the clinical relation between them.

In accordance with the aforementioned approach, EXCERF first identify the leaf nodes corresponding to the two given diseases i and j. Should they share the same node within the first level of O_d, the distance $\check{D}^{i,j}_d$ would be the length of the shortest path connecting these them, which equals to the minimum number of traversed edges to reach i from j. Conversely, $\check{D}^{i,j}_d$ would be a predetermined maximum distance md_d. Then the model calculates relation weight $\tilde{r}^{i,j}_d$ based on $\check{D}^{i,j}_d$ according to:

$$\tilde{r}^{i,j}_d = \ln(1/\check{D}^{i,j}_d) - \ln(1/md_d) \tag{3}$$

The calculation of $\tilde{r}^{i,j}_p$, which pertains to the relation weight of two given procedures, following the same strategy as follows:

$$\tilde{r}^{i,j}_p = \ln(1/\check{D}^{i,j}_p) - \ln(1/md_p) \tag{4}$$

Subsequently, let $\overline{H} = [\bar{h}^1, \bar{h}^2, \ldots, \bar{h}^n]$ denote the input sequence, where $\bar{h}^t \in \mathbb{R}^{dim}$ is the hidden representation of the tth entity. The standard self-attention applies three matrices $W_Q \in \mathbb{R}^{dim \times dim_Q}$, $W_K \in \mathbb{R}^{dim \times dim_K}$ and $W_V \in \mathbb{R}^{dim \times dim_V}$ to project $\overline{H}$ into queries Q of dimension dim_Q, keys K of dimension dim_K and values V of dimension dim_V as follows:

$$Q = \overline{H} W_Q,\ K = \overline{H} W_K,\ V = \overline{H} W_V \tag{5}$$

Q and K would determine affinities among entities according to:

$$E = \frac{QK^\intercal}{\sqrt{dim_K}} \tag{6}$$

which is then used to yield the output values $\widetilde{H}$ as follows:

$$\widetilde{H} = \text{Attention}(Q, K, V) = \text{softmax}(E)V \tag{7}$$

In order to incorporate clinical knowledge into attention computation, EXCERF introduces knowledge tuples into the calculation process. Specifically, for a pair of representations $\bar{h}^i_*$ and $\bar{h}^j_*$ of two entities within the input sequence, EXCERF identifies the corresponding knowledge tuple and embeds the relation weight $\tilde{r}_*$ with a trainable embedding matrix $W_{b,*}$ to yield the bias term $b^{i,j}_*$ as:

$$b^{i,j}_* = \tilde{r}^{i,j}_* W_{b,*} \tag{8}$$

Then the model incorporates it into the self-attention module as follows:

$$E^{i,j}_* = \frac{(\bar{h}^i_* W_{Q,*})(\bar{h}^j_* W_{K,*})^\intercal}{\sqrt{dim_K}} + b^{i,j}_* \tag{9}$$

The subsequent computation follows the standard self-attention mechanism. EXCERF applies the softmax function on E_* to derive the attention weight, and then read context from values V as follows:

$$\widetilde{H}_* = \text{softmax}(E_*)(\overline{H}_* W_{V,*}) \tag{10}$$

After that, the model applies a layer normalization to the combination of $\overline{H}_*$ and $\widetilde{H}_*$ according to:

$$\hat{H}_* = \text{LayerNorm}(\overline{H}_* + \widetilde{H}_*) \tag{11}$$

Following this process, EXCERF leverages KPSA to encode embeddings results $\overline{H}^t_d$ for diagnoses and $\overline{H}^t_p$ for procedures into $\hat{H}^t_d$ and $\hat{H}^t_p$ respectively.

3.4. External MemNN

In pursuit of effective extraction of cohort features, EXCERF builds an external key–value MemNN $\widetilde{M}_*$ universally accessible to the entire patient cohort, enabling patients to capture related category-specific features according to their personalized representations. Specifically, EXCERF uses two different memory units $\widetilde{M}_{*,key}$ and $\widetilde{M}_{*,value}$ that are implemented with two linear layers respectively as the key and value, then the model averages $\hat{H}_*^t$ to generate the query $\tilde{e}_*^t$, enabling the calculation of attention weights over the memory keys $\widetilde{M}_{*,key}$ as follows:

$$\tilde{a}_*^t = \text{softmax}((\tilde{e}_*^t)^\intercal \widetilde{M}_{*,key}) \tag{12}$$

Subsequently, the model retrieves context $\tilde{o}_*^t$ from memory values $\widetilde{M}_{*,value}$ based on attention weight as follows:

$$\tilde{o}_*^t = \tilde{a}_*^t \widetilde{M}_{*,value} \tag{13}$$

thereby extracting cohort features aligned with the group to which the current patient belongs. After that, the extracted features are integrated with the query $\tilde{e}_*^t$ to generate experience-enhanced representation e_*^t according to:

$$e_*^t = \tilde{e}_*^t + \tilde{o}_*^t \tag{14}$$

Incorporating shared features among patients via the external MemNN eliminates the need for explicit consideration of patient correlations. Thus, this approach alleviates computational burdens and prevents the use of irrelevant patient information arising from unreasonable correlation definitions.

3.5. Patient representation for recommendation

EXCERF further encodes EHRs stemming from all admissions of an individual patient to capture inherent temporal dependencies. EXCERF computes the average of e_d^t and e_p^t to generate e^t, which subsequently acts as the input for GRU to generate the admission representation q^t as follows:

$$q^t, h^t = GRU(h^{t-1}, e^t) \tag{15}$$

After obtaining $q^t (t \in [1, T])$, EXCERF build a local key–value MemNN M for each patient. The keys and values are historical admission representations q^t and prescriptions $m^t (t \in [1, T-1])$. The model uses q^T as initial query u^0 to calculate attention weight a^1 as follows:

$$a^1 = \text{softmax}((u^0)^\intercal M_{keys}) \tag{16}$$

a^1 is then used to read relevant medications from the values according to:

$$o^1 = a^1 s_m^* Z^1 \tag{17}$$

where s_m^* represents the multi-hot vectors of MemNN values $m^t (t \in [1, T-1])$, and Z^1 represents medication embeddings obtained by Graph Convolutional Networks (GCN) [35]. After that, the read result o^1 is combined with u^0 to update the representation as follows:

$$u^1 = u^0 + o^1 \tag{18}$$

Once EXCERF updates the initial query u^0 by incorporating medication information from M, the newly integrated data alters clinically meaningful associations between the query and the historical patient conditions in M. Therefore, EXCERF continues to read from M based on the current associations, ensuring that the model keeps acquiring historical medication information that is most relevant to the updated query. Specifically, during the kth hop of reading, EXCERF employs u^{k-1} as the query, and calculates the attention a^k between the query and the keys M_{keys} as follows:

$$a^k = \text{softmax}((u^{k-1})^\intercal M_{keys}) \tag{19}$$

Then EXCERF reads o^k from M based on a^k according to:

$$o^k = a^k s_m^* Z^k \tag{20}$$

and o^k is used to update u^{k-1} for the next hop of reading as follows:

$$u^k = u^{k-1} + o^k \tag{21}$$

For each hop of reading on the local MemNN, EXCERF applies GCN on the co-occurrence graph G_c to update medication embeddings Z^k as follows:

$$Z = \hat{A}\tanh(\hat{A}W_e^1)W_e^2 \tag{22}$$

where W_e^1 and W_e^2 are weight parameter matrices, and $\hat{A}$ is derived from the graph according to:

$$\hat{A} = \hat{D}^{-\frac{1}{2}}(A_c + I)\hat{D}^{-\frac{1}{2}} \tag{23}$$

Here, $A_c \in \mathbb{R}^{|S_m| \times |S_m|}$ is the adjacency matrix of G_c, I denotes an identity matrix, and $\hat{D}$ is a diagonal matrix with elements defined as $\hat{D}^{i,j} = \sum_j A_c^{i,j}$.

As EXCERF keeps updating Z^k throughout the multi-hop reading on the local MemNN, the model applies attention mechanism on the final medication embeddings $Z^{\mathbb{K}}$ after $\mathbb{K}$ hops of reading to obtain relevant medication information. To do so, EXCERF uses q^T as the query, and calculates the attention weight a^r according to:

$$a^r = \text{softmax}((q^T)^\intercal Z^{\mathbb{K}}) \tag{24}$$

a^r is then used to read relevant medication information from $Z^{\mathbb{K}}$ as follows:

$$r = a^r Z^{\mathbb{K}} \tag{25}$$

Finally, EXCERF combines the admission representation q^T, the read results $u^{\mathbb{K}}$ from the local MemNN, and the relevant medication information r from $Z^{\mathbb{K}}$ to generate recommendations. These elements are combined to execute the recommendation as follows:

$$\hat{y} = \sigma(f([q^T, u^{\mathbb{K}}, r])) \tag{26}$$

where $\sigma(\cdot)$ is a sigmoid function, $f(\cdot)$ is a fully connected layer, and $[\cdot]$ stands for concatenation. The result $\hat{y} \in \mathbb{R}^{|S_m|}$ is a probability vector indicating the likelihood of each drug being recommended, and those drugs with a probability exceeding 0.5 form the final prescription $\hat{Y}$ that defined as:

$$\hat{Y} = \{i | \hat{y}_i > 0.5, 1 \le i \le |S_m|\} \tag{27}$$

3.6. Training and inference

EXCERF minimizes a weighted loss $\mathcal{L}_r$ to optimize its parameters, and the loss is defined as follows:

$$\mathcal{L}_r = \alpha_1 \mathcal{L}_b + \alpha_2 \mathcal{L}_m \tag{28}$$

where $\mathcal{L}_b$ is a binary cross-entropy loss that is calculated as:

$$\mathcal{L}_b = -\sum_{i=1}^{|S_m|} [y_i \log \sigma(\hat{y}_i) + (1 - y_i) \log(1 - \sigma(\hat{y}_i))] \tag{29}$$

$\mathcal{L}_m$ is a multi-label margin loss that is calculated according to:

$$\mathcal{L}_m = \sum_{i=1}^{|S_m|} \sum_{j=1}^{|\hat{Y}|} \frac{\max(0, 1 - (\hat{y}[\hat{Y}_j] - \hat{y}_i))}{|S_m|} \tag{30}$$

These two loss functions are integrated with two weights α_1 and α_2 that follow the constraint $\alpha_1 + \alpha_2 = 1$. The variable y represents the ground truth medications, i.e., s_m^T, and $\hat{y}_i$ as well as $\hat{y}[\hat{Y}_i]$ stand for the probabilities that medications i and $\hat{Y}_i$ would be prescribed to the patient.

The training algorithm is detailed as follows:

Algorithm 1 Training EXCERF

Require: Training set X_R, training epochs $Epoch$, read number of hops $\mathbb{K}$ on local MemNN, loss weight α_1 and α_2;

calculate adjacency matrix A_c;
obtain knowledge tuples $\tilde{K}_d$ and $\tilde{K}_p$ using Eq. (3) and Eq. (4);
for $i = 1$ to $Epoch$ **do**
  for $j = 1$ to $|X_R|$ **do**
    get patient records X_j;
    for $t = 1$ to T_j **do**
      obtain medical embeddings $\overline{H}_d^t$ and $\overline{H}_p^t$ in Eq. (1) and Eq. (2);
      generate knowledge-powered embeddings $\widehat{H}_d^t$ and $\widehat{H}_p^t$ in Eq. (11);
      read cohort features $\tilde{o}_d^t$ and $\tilde{o}_p^t$ in Eq. (13);
      generate experience-enhanced e_d^t and e_p^t in Eq. (14);
      generate admission representations q^t in Eq. (15);
    end for
    build the local MemNN M;
    for $k = 1$ to $\mathbb{K}$ **do**
      read o^k from M in Eq. (17);
      obtain updated query u^k in Eq. (18);
    end for
    read relevant medication information r in Eq. (25);
    calculate medication prediction $\hat{y}^{(j)}$ in Eq. (26);
    update model parameters by optimizing loss in Eq. (28);
  end for
end for

Table 2
Statistics of the selected patients.

#patient	6350
#diagnoses	1958
#procedures	1430
#medications	151
average #visits	2.37
average #diagnoses per visit	13.63
average #procedures per visit	4.54
average #medications per visit	19.86
max #visits	29
max #diagnoses per visit	39
max #procedures per visit	32
max #medications per visit	55

4. Experiments

4.1. Experimental settings

Experiments are conducted on the real-world clinical dataset MIMIC-III [36] to compare EXCERF with different baseline methods. The model is implemented with PyTorch and trained on a NVIDIA TITAN Xp GPU.[1] To ensure a rigorous hyperparameter tuning process, Bayesian optimization using Gaussian Processes is employed. Additionally, 10-fold cross validation is used to assess the performance of EXCERF.

MIMIC-III is an authentic database of clinical data comprising EHRs associated with over 40,000 patients, and these records cover diverse aspects of patient conditions. In the context of this study, patients with a minimum of two hospital admissions are selected, and their diagnoses as well as procedures would serve as the inputs for EXCERF, while the medications prescribed within the initial 24 h of hospitalization are the ground truth for the recommendation. The statistics of the selected records are shown in Table 2.

[1] https://github.com/yanda-wang/EXCERF.

EXCERF is compared with the following baseline methods.

- **GAMENet** [11]: GAMENet learns longitudinal patient EHRs and writes the results into a key–value MemNN to emulate case-based retrievals inherent to clinical practice for the recommendation.
- **G-BERT** [15]: G-BERT uses GNN to model the hierarchical structures of medical entities, and applies the results to embed the inputs of a transformer-based pre-trained encoder for the recommendation.
- **Leap** [32]: Leap treats the recommendation as a multi-instance and multi-label learning challenge. The approach regards disease conditions as instances while medications act as the associated labels.
- **RETAIN** [37]: RETAIN introduces a two-layer attention model to detect influential historical visits and clinical variables, and attends to EHRs in a reverse chronological order to focus to the recent visits.
- **SARMR** [27]: SARMR adopts a key–value MemNN to store encoded historical records of patients, and then attends to the MemNN to enhance patient representations for medication recommendation.

We use multiple metrics to evaluate different methods, including Jaccard Similarity (Jaccard), Precision Recall AUC (PRAUC), Average Precision (Avg-P), Average Recall (Avg-R), Average F1 (F1), Average DDI rate (DDI) and Average Normalized Discounted Cumulative Gain (N@K). Jaccard is defined as the size of intersection divided by the size of union of ground truth medications $Y^{(i)}$ and predict medications $\hat{Y}^{(i)}$ as follows:

$$\text{Jaccard} = \frac{1}{\sum_i^N 1} \sum_i^N \frac{|Y^{(i)} \cap \hat{Y}^{(i)}|}{|Y^{(i)} \cup \hat{Y}^{(i)}|} \tag{31}$$

where N is the number of patients in test set. Precision and Recall of the ith patient are known as $\text{P}^{(i)} = |Y^{(i)} \cap \hat{Y}^{(i)}|/|\hat{Y}^{(i)}|$ and $\text{R}^{(i)} = |Y^{(i)} \cap \hat{Y}^{(i)}|/|Y^{(i)}|$, so Avg-P, Avg-R and F1 are calculated as follows:

$$\text{Avg-P} = \frac{1}{\sum_i^N 1} \sum_i^N \frac{|Y^{(i)} \cap \hat{Y}^{(i)}|}{|\hat{Y}^{(i)}|} \tag{32}$$

$$\text{Avg-R} = \frac{1}{\sum_i^N 1} \sum_i^N \frac{|Y^{(i)} \cap \hat{Y}^{(i)}|}{|Y^{(i)}|} \tag{33}$$

$$\text{F1} = \frac{1}{\sum_i^N 1} \sum_i^N \frac{2 \times \text{P}^{(i)} \times \text{R}^{(i)}}{\text{P}^{(i)} + \text{R}^{(i)}} \tag{34}$$

DDI refers to the phenomenon where interactions between drugs lead to adverse outcomes or side effects. Given a prescription $\hat{Y}^{(i)}$, DDI rate is defined as the proportion of medication pairs that lead to interaction, therefore the average DDI rate (DDI) is defined as follows:

$$\text{DDI} = \frac{1}{\sum_i^N 1} \frac{\sum_{\bar{i},\bar{j}} |\{(m_{\bar{i}}, m_{\bar{j}}) \in \hat{Y}^{(i)} | (m_{\bar{i}}, m_{\bar{j}}) \in \mathcal{E}_D\}|}{\sum_{\bar{i},\bar{j}} 1} \tag{35}$$

Here, $\mathcal{E}_D$ is the edge set of the DDI graph, where any two medications interact are connected. The average DDI rate across all patients is used as the metric.

NDCG@K is computed as the ratio of the Discounted Cumulative Gain at rank K (DCG@K) to the Ideal Discounted Cumulative Gain at rank K (IDCG@K). As DCG@K for the ith patient is defined as $\text{DCG@K}^{(i)} = \sum_{\bar{i}=1}^K \bar{r}_{\bar{i}}^{(i)} / \log_2(\bar{i}+1)$, where $\bar{r}_{\bar{i}}^{(i)}$ is the relevance score of the item at rank $\bar{i}$, and IDCG@K is defined as $\text{IDCG@K}^{(i)} = \sum_{\bar{i}=1}^K \bar{r}_{\bar{i}}^{ideal,(i)} / \log_2(\bar{i}+1)$, where $\bar{r}_{\bar{i}}^{ideal,(i)}$ represents the relevance score of items in the perfectly ranked order, the average NDCG@K is given by:

$$\text{N@K} = \frac{1}{\sum_i^N 1} \sum_i^N \frac{\text{DCG@K}^{(i)}}{\text{IDCG@K}^{(i)}} \tag{36}$$

Table 3
Performance comparison of different baseline methods.

Metric	GAMENet	G-BERT	Leap	RETAIN	SARMR	EXCERF
Jaccard	0.4987	0.4402	0.4481	0.4882	0.5035	0.5136
PRAUC	0.7585	0.6812	0.6467	0.7500	0.7619	0.7716
Avg-P	0.5663	0.5738	0.6428	0.6129	0.5724	0.5819
Avg-R	0.8187	0.6702	0.6173	0.7320	0.8158	0.8221
F1	0.6558	0.6001	0.6105	0.6480	0.6601	0.6698
DDI	0.3920	0.4113	0.4110	0.4143	0.3785	0.3977
N@5	0.8977	0.8161	0.7472	0.9026	0.8927	0.9054
N@10	0.8496	0.7572	0.7168	0.8528	0.8492	0.8600
N@15	0.8151	0.7314	0.7043	0.8091	0.8179	0.8245
N@20	0.8037	0.7479	0.7091	0.7914	0.8064	0.8130

4.2. Comparison with baseline methods

Table 3 presents the comparison between EXCERF and different baseline methods, demonstrating the superior performance of EXCERF across Jaccard, PRAUC, F1, and N@K at varying values of K. Despite not explicitly leveraging DDI knowledge during prescription, EXCERF achieves a DDI rate comparable to GAMENet that places significant emphasis on reducing DDI, and it is the only method with a DDI rate below 0.4 without direct reliance on DDI knowledge. While EXCERF does not obtain the highest Avg-P, it shows the best F1 score and demonstrates notable advantages. Among those baseline methods, G-BERT exhibits the lowest values in F1 and Jaccard. Although G-BERT leverages clinical knowledge when encoding EHRs, it treats this knowledge merely as auxiliary information to construct the inputs of the encoder, rather than integrating it to enhance connections between clinically related entities, resulting in suboptimal representations that fail to accurately describe patient conditions. Similarly, Leap performs worse than EXCERF in terms of most metrics, since it neglects the involvement of explicit clinical knowledge, relying solely on semantic relationships among entities to construct patient representations. While semantic relationships provide insight into clinical connections, they fail to compete with explicit integration of clinical knowledge, leading to less informative representations of patient conditions. In contrast, GAMENet and SARMR achieves performance closer to EXCERF across multiple metrics. since they apply a similar strategy that constructs a local MemNN to reference historical admissions. However, they do not take cohort features from patients with similar conditions into consideration, and the limited records of individual patients do not provide sufficient description of their health conditions, preventing GAMENet and SARMR from matching the performance of EXCERF.

4.3. Robustness and genericity

To assess the robustness and genericity of EXCERF across different health conditions, we evaluate its performance on nine diseases groups related to major organs and human systems based on patients in the test dataset that affected by these diseases, and summarize the results in Table 4. As the table shows, EXCERF demonstrates well-balanced performance across all diseases, highlighting its robustness and genericity to diverse health conditions. The model achieves an F1 that is higher than 0.67 for most patient cohorts, peaking at 0.6931 on patients with septicemia, a severe and life-threatening infection, while only patients suffering from stomach-related diseases show an F1 below 0.65. Moreover, although EXCERF is not explicitly designed for DDI reduction, it still maintains noteworthy performance on DDI rate to maintain the safety of generated recommendations, and only the cohort with lung-related diseases exhibits a DDI rate higher than 0.4.

EXCERF leverages historical admissions as a critical reference for medication recommendation, but the first admission of a patient lacks the reference with the absence of historical admissions. Meanwhile, cohort features derived from other patients with similar conditions provide an alternative source of reference to overcome the problem. To demonstrate the adverse effects of missing historical admissions and the important role of cohort features, we summarize the performance of EXCERF under these conditions in Table 5. As the table shows, the absence of cohort features leads to poorer performance for the first admission compared to subsequence admissions, where historical records are available. Meanwhile, leveraging cohort features assists in achieving substantial improvement in performance for the first admission. Moreover, when historical admissions are available, cohort features further contribute to additional increase in performance. Note that despite the incorporation of cohort features, a performance gap still exists between the first admission and subsequent ones, and it is worthwhile to further explore ways to cope with missing historical records.

4.4. Case study

We select a patient from the test dataset and generate recommendations to compare EXCERF with different baseline methods, the results for two separate admissions are presented in Tables 6 and 7. As shown in the tables, EXCERF demonstrates overall superior performance, and achieves 14 correct predictions for both admissions. G-BERT and RETAIN lack the ability to generate medications for the first admission. Although G-BERT avoids generating wrong medications for the second admission, it misses nearly half of the medications, and RETAIN exhibits the poorest performance with 7 wrong predictions, a result matched by Leap. For the first admission, where no historical records are available for reference, GAMENet, Leap, and SARMR either miss many of the ground truth medications or generate a significant number of wrong ones. In contrast, EXCERF leverages cohort features from patients with similar conditions, which enables it to outperform these baseline methods. For the second admission, all methods show improved performance compared to the first admission. EXCERF maintains its advantage and shows the best overall results. The enhanced connections between medical entities by clinical knowledge helps avoid missing closely related medications or generating irrelevant ones. Cohort features enable the model to consider all essential and diverse medications, generating prescriptions that contain multiple categories while minimizing omissions.

4.5. Ablation study

To demonstrate the necessity of various components within EXCERF, we eliminate or substitute these components one at a time to construct different variants. Then we compare EXCERF with these variants regarding their performance on medication recommendation. Moreover, to evaluate the computational demands of these components, the Average Training Time per Epoch (ATTE) is used as a metric during the comparative analysis.

- **No-KPSA:** No-KPSA removes the KPSA module with embedding matrices to generate inputs for the subsequent modules.
- **No-External-MemNN (No-Ext-Mem):** No-Ext-Mem removes the external MemNN component and focuses on personalized features of each patient without relying on clinical experience
- **No-Admission-Position (No-Adm-Pos):** No-Adm-Pos leaves out GRU overlook the temporal information expressed by admission positions when encoding clinical records.
- **No-Local-MemNN (No-Loc-Mem):** No-Loc-Mem removes the local MemNN component and relies on temporal patterns of multiple admissions alone to conduct the recommendation.
- **Single-Hop:** Single-hop conducts a single hop of reading on the local MemNN without attending different admissions based on the updated query for contextual information.

Table 4
Robustness and genericity of EXCERF across diverse disease categories.

Metric	Cerebrovascular	Diabetes	Heart	Hypertension	Liver	Lung	Pancreas	Septicemia	Stomach
Jaccard	0.5303	0.5276	0.5391	0.5192	0.5046	0.5294	0.5174	0.5374	0.4901
PRAUC	0.7944	0.7791	0.7931	0.7746	0.7602	0.7865	0.7637	0.7938	0.7655
Avg-P	0.6016	0.6002	0.6108	0.5905	0.5717	0.6006	0.5857	0.6130	0.5567
Avg-R	0.8315	0.8223	0.8261	0.8183	0.8220	0.8264	0.8201	0.8257	0.8189
F1	0.6847	0.6845	0.6924	0.6752	0.6598	0.6856	0.6754	0.6931	0.6490
DDI	0.3892	0.3793	0.3918	0.3927	0.3540	0.4044	0.3817	0.3718	0.3866
N@5	0.9234	0.9072	0.9241	0.9082	0.9069	0.9228	0.9163	0.9286	0.9118
N@10	0.8869	0.8746	0.8842	0.8661	0.8449	0.8856	0.8658	0.8954	0.8448
N@15	0.8527	0.8370	0.8481	0.8313	0.8109	0.8451	0.8293	0.8570	0.8114
N@20	0.8366	0.8141	0.8299	0.8148	0.8103	0.8194	0.8028	0.8314	0.8148

Table 5
Robustness and genericity to missing data.

Metric	No cohort features		With cohort features	
	No historical admissions	With historical admissions	No historical admissions	With historical admissions
Jaccard	0.5034	0.5169	0.5052	0.5206
PRAUC	0.7623	0.7726	0.7678	0.7748
Avg-P	0.5745	0.5846	0.5741	0.5884
Avg-R	0.8114	0.8245	0.8172	0.8262
F1	0.6606	0.6728	0.6622	0.6761
DDI	0.4020	0.3977	0.4000	0.3958
N@5	0.8993	0.9043	0.9025	0.9078
N@10	0.8516	0.8629	0.8560	0.8635
N@15	0.8168	0.8278	0.8219	0.8265
N@20	0.8098	0.8128	0.8128	0.8128

Table 6
Example of recommendation for the first admission of a patient with two.

Method	Results
GroundTruth	A01A, A02B, N01A, C07A, N02B, A12C, B01A, C09A, C10A, C08C, A06A, C02D, A12B, N06A, N05C, H03B, N03A (17)
GAMENet	**Correct**: A02B, N01A, C09A, A06A, A12B, C07A, N02B, B01A, A12C, C02D, A01A, C10A, N03A (13) **Missed**: C08C, N06A, N05A, H03B (4) **Wrong**: B05C, A12A, A07A, A10A, C03C, N02A, J01D, A04A, H04A (9)
G-BERT	**Correct**:/(0) **Missed**: A01A, A02B, N01A, C07A, N02B, A12C, B01A, C09A, C10A, C08C, A06A, C02D, A12B, N06A, N05C, H03B, N03A (17) **Wrong**:/(0)
Leap	**Correct**: A02B, A12B, A06A, B01A, N02B, A12C, C07A, C10A, N03A (9) **Missed**: A01A, N01A, C09A, C08C, C02D, N06A, N05A, H03B (8) **Wrong**: B05C, A07A, J01D, H04A, C09C (5)
RETAIN	**Correct**:/(0) **Missed**: A01A, A02B, N01A, C07A, N02B, A12C, B01A, C09A, C10A, C08C, A06A, C02D, A12B, N06A, N05C, H03B, N03A (17) **Wrong**:/(0)
SARMR	**Correct**: N01A, A02B, A01A, C02D, C07A, A06A, N05A, N02B, C08C, B01A, A12C, C09A, C10A, N03A, A12B (15) **Missed**: N06A, H03B (2) **Wrong**: B05C, A12A, C01C, A07A, C03C, N02A, C01B, N05C, J01D, A04A, H04A (11)
EXCERF	**Correct**: A02B, N01A, C09A, A06A, A12B, C08C, C07A, N02B, B01A, A12C, C02D, A01A, C10A, N03A (14) **Missed**: N06A, N05A, H03B (3) **Wrong**: B05C, C01C, A07A, N05C, J01D, H04A (6)

4.5.1. Quantitative comparison

The comparative results are summarized in Table 8. As the table shows, EXCERF achieves the highest Jaccard, PRAUC, F1 and N@K at varying values of K. Although EXCERF falls slightly short of No-Loc-Mem on Avg-P, it demonstrates balanced optimization of Avg-P and Avg-R that leads to the highest F1, while No-Loc-Mem exhibits the lowest Avg-R. This difference can likely be attributed to the distribution of attention weights over MemNN slots. When historical admissions are not highly relevant to the current one, attention weights still spread across different MemNN slots eventually, potentially introducing irrelevant medications. No-Loc-Mem avoids the potential inference, but this comes at the cost of the lowest recall, resulting in suboptimal performance compared to EXCERF. Meanwhile, No-KPSA exhibits substantial decreases in all metrics, especially the decline of Jaccard from 0.5136 to lowest value 0.5080. This demonstrates the importance of KPSA in integrating clinical knowledge into self-attention mechanism, which facilitate information transfer to construct informative representations for the recommendation. No-Ext-Mem also exhibits reduced performance across different metrics compared to EXCERF, which highlights the significance of the external MemNN that learns cohort features. The absence of such cohort features limits No-Ext-MemN to relying solely on personalized conditions, resulting in inferior representations for the recommendation. Furthermore, the performance of No-Adm-Pos

Table 7
Example of recommendation for the second admission of a patient with two.

Method	Results
GroundTruth	A01A, A02B, A12C, A07A, C07A, N02B, B01A, C09A, C10A, C08C, A06A, C02D, J01D, B05C, N06A, N03A, N05A, H03B (18)
GAMENet	**Correct**: A02B, C09A, A06A, C08C, C07A, N02B, B01A, A12C, N05A, N06A, C02D, B05C, A01A, C10A, N03A (15) **Missed**: A07A, J01D, H03B (3) **Wrong**: A12A, A12B, N02A, A04A (4)
G-BERT	**Correct**: A12C, A02B, B01A, C02D, B05C, N02B, C07A, A01A, C09A, C10A, A06A (11) **Missed**: C08C, N06A, H03B, A07A, J01D, N03A, N05A (7) **Wrong**:/(0)
Leap	**Correct**: A02B, C09A, A06A, B01A, N02B, A12C, N05A, C02D, B05C, A01A, C07A, A07A, N03A (13) **Missed**: C10A, C08C, J01D, N06A, H03B (5) **Wrong**: C01C, A10A, N01A, A04A, N05C, N02A, A12B (7)
RETAIN	**Correct**: A02B, C09A, A06A, C07A, N02B, B01A, A12C, N05A, N06A, C02D, B05C, A01A, C10A, A07A, N03A (15) **Missed**: C08C, J01D, H03B (3) **Wrong**: A12A, A10A, N01A, C03C, A12B, N02A, N05C (7)
SARMR	**Correct**: B05C, A02B, A01A, C02D, C07A, A06A, N05A, A07A, C10A, N02B, J01D, C08C, B01A, C09A, A12C (15) **Missed**: N06A, N03A, H03B (3) **Wrong**: A12B, N02A, A02A, J01M, A04A, H04A (6)
EXCERF	**Correct**: A02B, C09A, A06A, C08C, C07A, N02B, B01A, A12C, C02D, B05C, A01A, C10A, A07A, N03A (14) **Missed**: J01D, N06A, N05A, H03B (4) **Wrong**: A12B, N02A (2)

Table 8
Performance comparison of different ablation variants.

Metric	No-KPSA	No-Ext-Mem	No-Adm-Pos	No-Loc-Mem	Single-Hop	EXCERF
Jaccard	0.5080	0.5107	0.5087	0.5115	0.5080	0.5136
PRAUC	0.7710	0.7678	0.7642	0.7643	0.7658	0.7716
Avg-P	0.5772	0.5801	0.5798	0.5846	0.5776	0.5819
Avg-R	0.8181	0.8184	0.8143	0.8126	0.8166	0.8221
F1	0.6647	0.6672	0.6657	0.6680	0.6649	0.6698
DDI	0.3972	0.3996	0.3973	0.3929	0.3897	0.3977
N@5	0.9051	0.9020	0.9010	0.8920	0.9022	0.9054
N@10	0.8581	0.8576	0.8525	0.8514	0.8561	0.8600
N@15	0.8240	0.8228	0.8197	0.8179	0.8200	0.8245
N@20	0.8128	0.8115	0.8095	0.8072	0.8090	0.8130
ATTE	64.9266	292.1382	285.4111	261.4657	262.3249	292.9128

drops marginally across various metrics, which indicates the necessity of temporal dependencies in medication recommendation. By removing temporal dependencies about admissions, No-Adm-Pos fails to capture personalized trends inherent in historical records, hindering its ability to predict preferences for subsequent medications. The absence of multi-hop reading on the local MemNN results in the degraded performance of Single-Hop. In contrast, continuous reading allows EXCERF to keep inferring preferred drugs based on the latest health conditions that influenced by previous prescriptions. While neither EXCERF nor its ablation variants directly focus on DDI reduction, all these methods maintain DDI rate within an acceptable range and balance safety with recommendation performance. Notably, No-Ext-Mem exhibits the highest DDI rate, which demonstrates that integrating cohort features into personalized representations aligns patient preferences with commonly prescribed drugs within the cohort, thereby reducing DDI rates related to personalized drugs when only individual conditions are considered.

EXCERF incurs the highest computational cost with an ATTE, suggesting that its advanced architecture may present scalability challenges. Specifically, No-KPSA exhibits the most significant decrease in ATTE compared to EXCERF, indicating that the transformer-based KPSA module demands substantial computational cost due to its clinically informed attention mechanism. By contrast, all the other variants show much smaller difference in ATTE compared to EXCERF. Notably, No-Ext-Mem has the ATTE value that nearly the same to that of EXCERF, while the external MemNN plays a crucial role in capturing cohort features, therefore enhancing performance with little computational cost. Moreover, the minimal difference in ATTE between No-Adm-Pos and EXCERF except for No-Ext-Mem suggests that sequential encoding of multiple admissions imposes a relatively modest computational burden while capturing temporal dependencies essential for patient representation. Both No-Loc-Mem and Single-Hop exhibit considerable reduction in ATTE that second only to No-KPSA, which highlights the considerable computational cost related to multi-hop reading on the local MemNN. In light of these findings, optimizing transformer-based calculation in KPSA and exploring more efficient alternatives to refer to historical prescriptions warrant further investigation to enhance the scalability and computational efficiency of EXCERF.

4.5.2. Clustering analysis

The comparison between EXCERF and ablation variants is further extended to clustering analysis. The analysis applies ICD codes of diseases and ATC codes of drugs to construct the vector $\ddot{q}$ of each patient for clustering. These codes are widely used standards in medical practice, with ICD codes classifying diseases based on etiology and anatomical locations, and ATC codes categorizing drugs into groups based on their therapeutic and pharmacological properties. Therefore, clustering based on ICD and ATC codes captures the inherent relationships between diseases and drugs as defined in clinical practice, facilitating the generation of medically meaningful groups.

The vector $\ddot{q} = [\ddot{q}_d, \ddot{q}_m]$ contains $\ddot{q}_d \in \mathbb{R}^{\ddot{S}_d}$ that describes the patient's diagnoses d^T and $\ddot{q}_m \in \mathbb{R}^{\ddot{S}_m}$ that characterizes medications $\hat{Y}$ generated

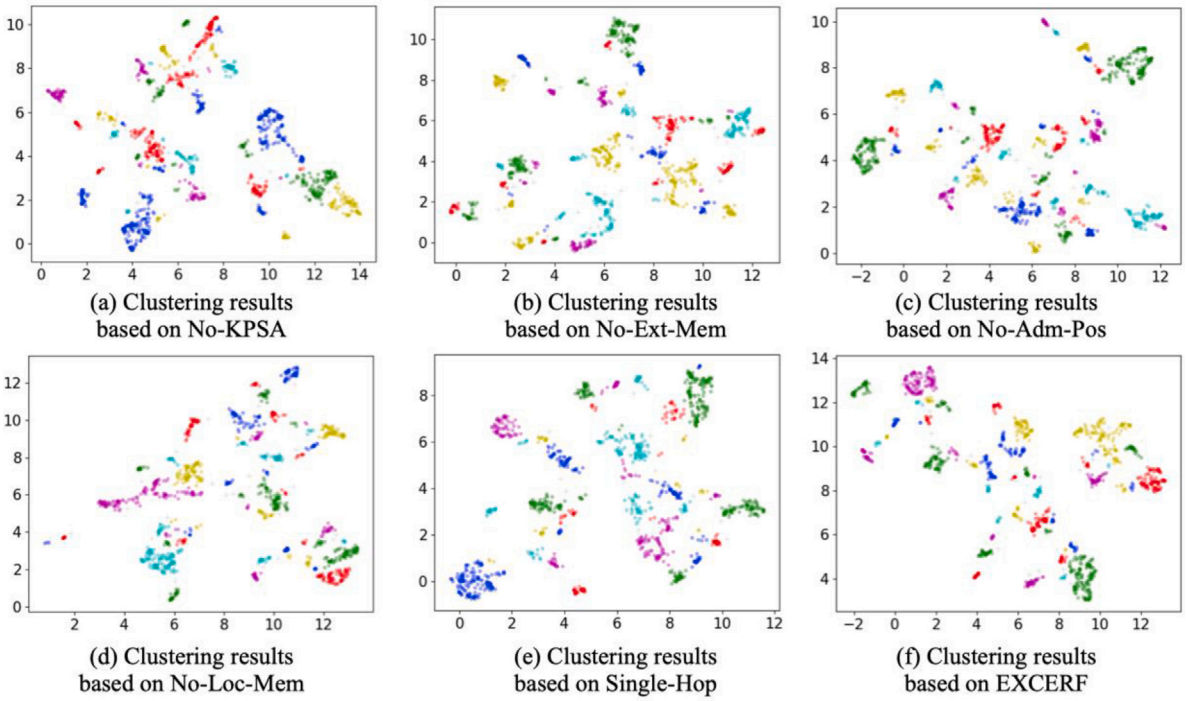


Fig. 2. Clustering results based on diseases and medications recommended by EXCERF and different variants.

by a model, where $\ddot{S}_d$ and $\ddot{S}_m$ correspond to the total number of disease categories and medication categories at the second level of the ICD and ATC code systems, respectively. The construction involves building a vector $\ddot{d} \in \mathbb{R}^{\ddot{S}_d}$, whose element $\ddot{d}_i$ denotes the count of diseases in d^T that belong to the ith category. The proportion of $\ddot{d}_i$ to the total number of diseases in the ith category forms the corresponding element in $\ddot{q}_d$. Following the same strategy, $\ddot{q}_m$ is constructed, and it is concatenated with $\ddot{q}_d$ to form $\ddot{q}$. Finally, Uniform Manifold Approximation and Projection for Dimension Reduction (UMAP) is applied to reduce the dimensions of $\ddot{q}$, and Density-Based Spatial Clustering of Applications with noise (DBSCAN) is used to conduct clustering.

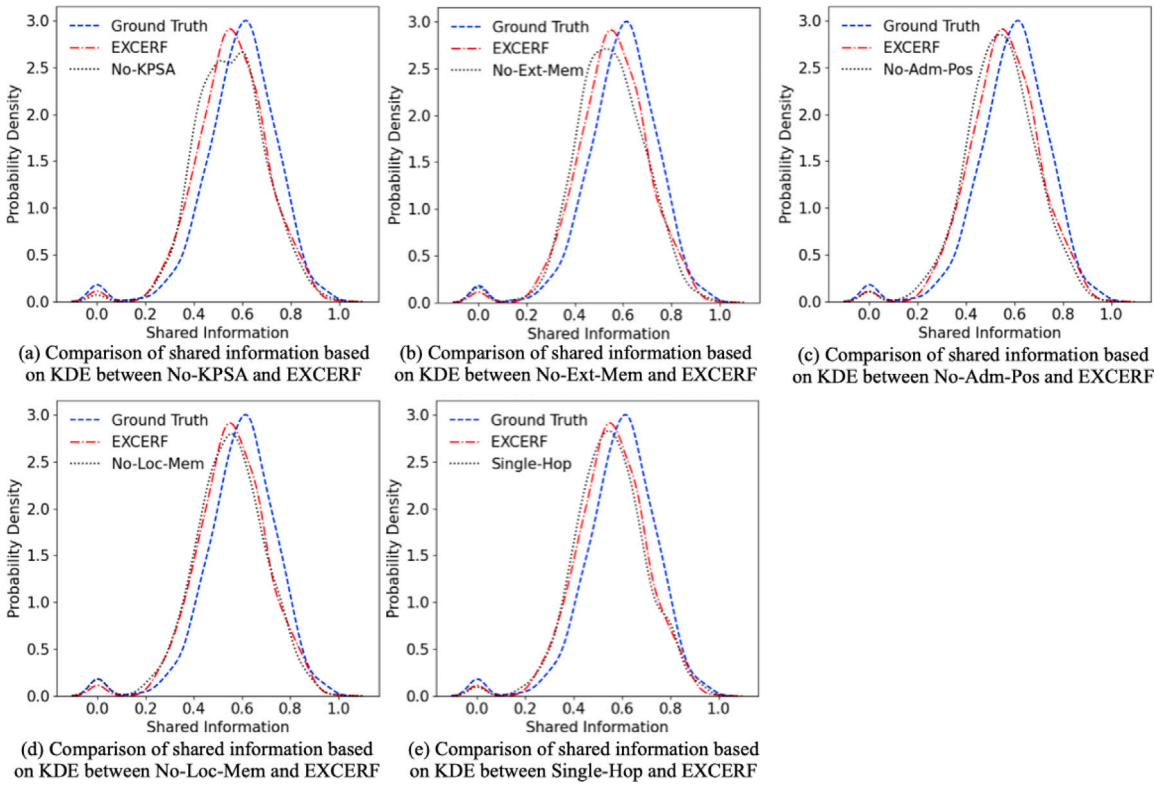
The clustering results are presented in Fig. 2. Individual points on graphs represent patients, and distinct colors are assigned to distinguish patients within various clusters. To accommodate the numerous clusters, these colors are cyclically used. Therefore, points with the same color yet being distant from each other belong to different clusters. Fig. 2 clearly delineates distinct clustering results between EXCERF and its different variants, revealing their disparate understandings of patient conditions and cohort features. To evaluate the quality of clusters, a metric based on shared medications among patients and corresponding neighboring points within the same cluster is proposed. Specifically, given a data point v representing a patient and its $\overline{N}$ neighboring points, their recommended medications are denoted by second level ATC codes, forming the set $\mathbb{Y}^v$ and $\mathbb{Y}^n (n \in [1, \overline{N}])$. Then we extract the interaction $\mathbb{Y}^v \cap \mathbb{Y}^n$, and define the shared information value as the proportion of $\mathbb{Y}^v \cap \mathbb{Y}^n$ to $|\mathbb{Y}^n|$. Then the average shared information value $\mathbb{S}^v$ for data point v and its $\overline{N}$ neighboring points is calculated as follows:

$$\mathbb{S}^v = \frac{\sum_{n=1}^{\overline{N}} |\mathbb{Y}^v \cap \mathbb{Y}^n| / |\mathbb{Y}^v|}{\overline{N}} \tag{37}$$

After calculating the average shared information values for all data points associated with clustering results of EXCERF and different variants, we perform Kernel Density Estimation (KDE) on these values to gain deep insights into cluster quality, and the results are illustrated in Fig. 3. The figure indicates that the curve of EXCERF is shifted to the right when compared to those of different variants, and the curve is also in closer proximity to the one corresponding to ground truth medications. The comparison shows that data points within clusters related to EXCERF demonstrate higher shared information values, which signifies increased similarity between patients within the same cluster. Moreover, the diminished separation between curves of EXCERF and ground truth, when compared to the larger distances between different variants and ground truth, proves that EXCERF generates prescriptions align with ground truth closely.

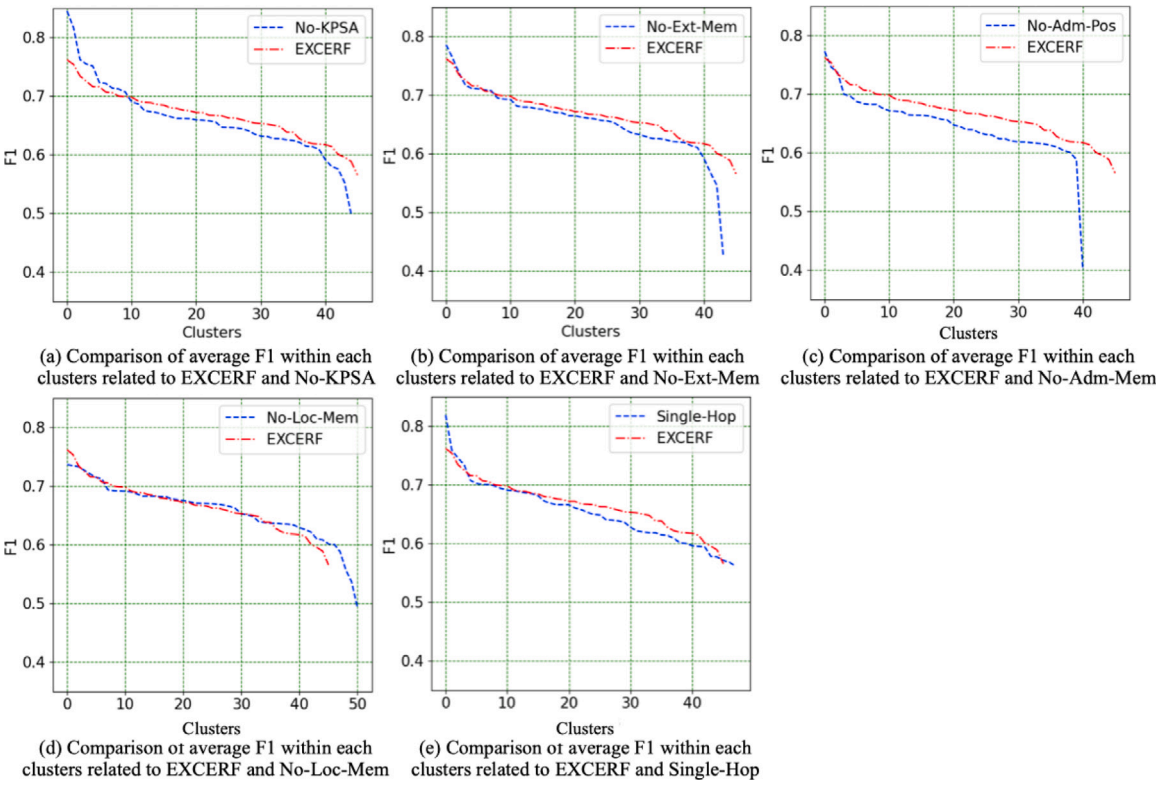
Subsequently, we evaluate the performance of EXCERF over different clusters via the average F1 values of patients within each cluster. These values are then sorted in descending order and compared with those of different variants. The comparative results are illustrated in Fig. 4. As the figure demonstrates, EXCERF achieves superior average F1 values across a broader spectrum of clusters compared to different variants. In contrast, the curve of No-KPSA reveals that it fails to compete with EXCERF due to the absence of clinical knowledge. Nevertheless, the curve exhibits a relatively gradual decline, indicating a degree of generalization across multiple clusters facilitated by the incorporation of cohort features. Although some peak average F1 values of No-Ext-Mem present roughly the same performance as EXCERF does, the majority of clusters associated with the variant exhibit decreased values compared to those of EXCERF. Also, while the curve for EXCERF shows a steady decline, the curve for No-Ext-Mem experiences a significant decrease towards the end, indicating that No-Ext-Mem does not generalize as effectively as EXCERF, as it cannot recommend effective prescriptions for certain patient cohorts. Meanwhile, No-Adm-Pos considers both clinical knowledge and cohort features, but the variant overlooks temporal patterns within multiple admissions of patients, explaining its inferior performance across the majority of clusters. No-Loc-Mem shows comparable performance to EXCERF for most of the clusters, but it exhibits a sharp decline towards the end, which highlights its inability to refer to historical records, resulting in suboptimal performance for certain clusters. By contrast, Single-Hop presents performance closest to EXCERF even with insufficient references to historical prescriptions, but a single hop of reading on local MemNN still hinders its ability to consistently extract related information from historical EHRs.

In addition to evaluating average F1 within clusters, the comparative analysis also extends to average Jaccard and PRAUC following the



(a) Comparison of shared information based on KDE between No-KPSA and EXCERF
(b) Comparison of shared information based on KDE between No-Ext-Mem and EXCERF
(c) Comparison of shared information based on KDE between No-Adm-Pos and EXCERF
(d) Comparison of shared information based on KDE between No-Loc-Mem and EXCERF
(e) Comparison of shared information based on KDE between Single-Hop and EXCERF

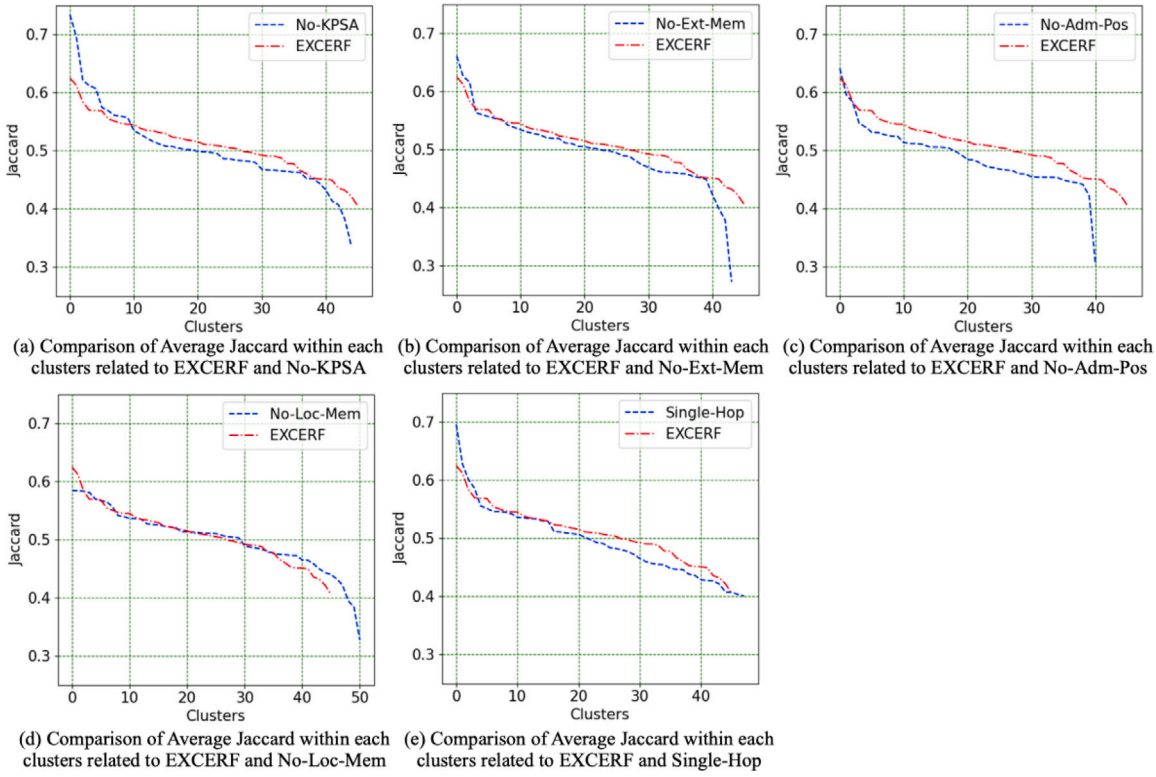
Fig. 3. Comparison of shared information between EXCERF and ablation variants.



(a) Comparison of average F1 within each clusters related to EXCERF and No-KPSA
(b) Comparison of average F1 within each clusters related to EXCERF and No-Ext-Mem
(c) Comparison of average F1 within each clusters related to EXCERF and No-Adm-Mem
(d) Comparison of average F1 within each clusters related to EXCERF and No-Loc-Mem
(e) Comparison of average F1 within each clusters related to EXCERF and Single-Hop

Fig. 4. Comparison of average F1 within clusters between EXCERF and variants.

same strategy. The results are illustrated in Figs. 5 and 6 respectively. It is clear that these figures present roughly the same comparison outcomes. EXCLEF outperforms all its variants across most clusters in Jaccard and PRAUC metrics, which further verifies the effectiveness of EXCLEF as well as its generalization capacity to diverse types of patients.



(a) Comparison of Average Jaccard within each clusters related to EXCERF and No-KPSA

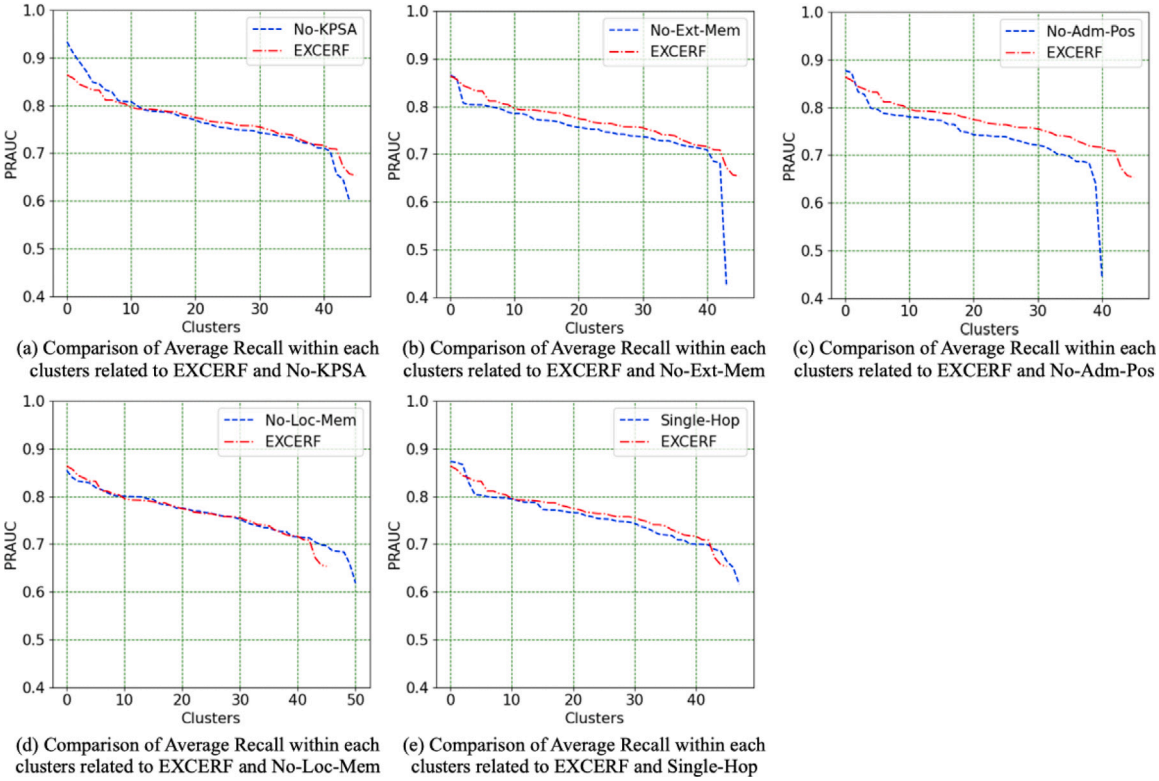
(b) Comparison of Average Jaccard within each clusters related to EXCERF and No-Ext-Mem

(c) Comparison of Average Jaccard within each clusters related to EXCERF and No-Adm-Pos

(d) Comparison of Average Jaccard within each clusters related to EXCERF and No-Loc-Mem

(e) Comparison of Average Jaccard within each clusters related to EXCERF and Single-Hop

Fig. 5. Comparison of average Jaccard within clusters between EXCERF and variants.



(a) Comparison of Average Recall within each clusters related to EXCERF and No-KPSA

(b) Comparison of Average Recall within each clusters related to EXCERF and No-Ext-Mem

(c) Comparison of Average Recall within each clusters related to EXCERF and No-Adm-Pos

(d) Comparison of Average Recall within each clusters related to EXCERF and No-Loc-Mem

(e) Comparison of Average Recall within each clusters related to EXCERF and Single-Hop

Fig. 6. Comparison of average PRAUC within clusters between EXCERF and variants.

5. Conclusion

In this work, we propose a novel medication recommendation model named EXCERF, which aims to address the shortcomings of existing methods in sufficient application of clinical knowledge and experience. EXCERF firstly identifies clinically meaningful associations among diverse medical entities within the clinical code systems, and integrates the knowledge as tuples into the self-attention mechanism to guide information exchange among corresponding entities, resulting in a simultaneous consideration of clinical knowledge and semantic correlations. Moreover, EXCERF implements a key–value MemNN shared across the entire patient cohort to acquire cohort features related to clusters of similar patients as clinical experiences, enabling patient to access associated cohort features through the attention mechanism without incurring exponential computational costs. Experiments conducted on real-world EHRs show that EXCERF outperforms all baseline methods, demonstrating its superior performance in medication recommendation and satisfactory generalization capabilities across diverse patient groups.

As we notice the high computational costs incurred by KPSA and the multi-hop reading on local MemNN, we expect to optimize the transformer-based calculation and explore more efficient alternatives to refer to historical prescriptions. Moreover, due to sparsity of available real-world clinical data and privacy protection policies, all experiments are conducted on MIMIC-III alone. We acknowledge that the results are influenced by hyperparameter tuning tailored to the specific dataset. Therefore, extending experiments to other clinical datasets and exploring adaptive tuning strategies to enhance genericity across diverse clinical contexts should be critical directions for future work. Additionally, the absence of manual evaluation by experts is another limitation, and feedback from domain experts would confirm the practical utility of the recommendations and verify that the clustering results align with real-world clinical practices. Addressing this limitation via collaboration with medical experts emerges as a priority in future research.

CRediT authorship contribution statement

Yanda Wang: Writing – original draft, Visualization, Validation, Software, Methodology, Data curation. **Weitong Chen:** Writing – review & editing, Supervision, Resources, Methodology, Conceptualization. **Lin Yue:** Writing – review & editing, Supervision, Project administration. **Ian Nabney:** Visualization, Validation, Formal analysis. **Dechang Pi:** Writing – review & editing, Conceptualization.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work the authors used ChatGPT in order to improve language and readability. After using this tool/service, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

The authors do not have permission to share data.

References

[1] C. Lyu, C. Qiu, K. Han, S. Li, V.S. Sheng, H. Rong, Y. Song, Y. Liu, Z. Liu, Automatic medical report generation combining contrastive learning and feature difference, Knowl.-Based Syst. 305 (2024) 112630.
[2] C.U.O. Kumar, S. Gajendran, B. Murugeshan, S. Marappan, K. R, EHR privacy preservation using federated learning with DQRE-scnet for healthcare application domains, Knowl.-Based Syst. 275 (2023) 110638.
[3] M. Sun, X. Yang, J. Niu, Y. Gu, C. Wang, W. Zhang, A cross-modal clinical prediction system for intensive care unit patient outcome, Knowl.-Based Syst. 283 (2024) 111160.
[4] P. Yan, M. Li, J. Zhang, G. Li, Y. Jiang, H. Luo, Cold SegDiffusion: A novel diffusion model for medical image segmentation, Knowl.-Based Syst. 301 (2024) 112350.
[5] T. Kim, J. Heo, H. Kim, K. Shin, S. Kim, VITA: 'carefully chosen and weighted less' is better in medication recommendation, in: Thirty-Eighth AAAI Conference on Artificial Intelligence, Vancourver, Canada, 2024, pp. 8600–8607.
[6] X. Li, S. Liang, Y. Hou, T. Ma, StratMed: Relevance stratification between biomedical entities for sparsity on medication recommendation, Knowl.-Based Syst. 284 (2024) 111239.
[7] Q. Chen, X. Li, K. Geng, M. Wang, Context-aware safe medication recommendations with molecular graph and DDI graph embedding, in: Thirty-Seventh AAAI Conference on Artificial Intelligence, Washington, DC, USA, 2023, pp. 7053–7060.
[8] E. Choi, M.T. Bahadori, A. Schuetz, W.F. Stewart, J. Sun, Doctor AI: predicting clinical events via recurrent neural networks, in: The 1st Machine Learning for Healthcare Conference, JMLR.org, Los Angeles, USA, 2016, pp. 301–318.
[9] R. Wu, Z. Qiu, J. Jiang, G. Qi, X. Wu, Conditional generation net for medication recommendation, in: Proceedings of the ACM Web Conference, Lyon, France, 2022, pp. 935–945.
[10] Y. Wang, W. Chen, D. Pi, L. Yue, M. Xu, X. Li, Multi-hop reading on memory neural network with selective coverage for medication recommendation, in: The 30th ACM International Conference on Information and Knowledge Management, Virtual Conference, 2021, pp. 2020–2029.
[11] J. Shang, C. Xiao, T. Ma, H. Li, J. Sun, Gamenet: Graph augmented memory networks for recommending medication combination, in: Thirty-Third AAAI Conference on Artificial Intelligence, Honolulu, Hawaii, USA, 2019, pp. 1126–1133.
[12] Y. Wang, W. Chen, D. Pi, L. Yue, Adversarially regularized medication recommendation model with multi-hop memory network, Knowl. Inf. Syst. 63 (1) (2021) 125–142.
[13] Y. Su, Y. Shi, W. Lee, L. Cheng, H. Guo, Tahdnet: Time-aware hierarchical dependency network for medication recommendation, J. Biomed. Informatics 129 (2022) 104069.
[14] Y. Wang, W. Chen, D. Pi, R. Boots, Graph augmented triplet architecture for fine-grained patient similarity, World Wide Web 23 (5) (2020) 2739–2752.
[15] J. Shang, T. Ma, C. Xiao, J. Sun, Pre-training of graph augmented transformers for medication recommendation, in: Proceddings of the Twenty-Eighth International Joint Conference on Artificial Intelligence, Macao, China, 2019, pp. 5953–5959.
[16] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A.N. Gomez, L. Kaiser, I. Polosukhin, Attention is all you need, in: Advances in Neural Information Processing Systems, Long Beach, CA, USA, 2017, pp. 5998–6008.
[17] S. Hong, J. Jang, H. Kwon, Enhancing performance of transformer-based models in natural language understanding through word importance embedding, Knowl.-Based Syst. 304 (2024) 112404.
[18] M. Zaheer, G. Guruganesh, K.A. Dubey, J. Ainslie, C. Alberti, S. Ontañón, P. Pham, A. Ravula, Q. Wang, L. Yang, A. Ahmed, Big bird: Transformers for longer sequences, in: Advances in Neural Information Processing Systems, Virtual Event, 2020, pp. 17283–17297.
[19] N. Kitaev, L. Kaiser, A. Levskaya, Reformer: The efficient transformer, in: 8th International Conference on Learning Representations, virtual event, 2020, pp. 1–12.
[20] A. Roy, M. Saffar, A. Vaswani, D. Grangier, Efficient content-based sparse attention with routing transformers, Trans. Assoc. Comput. Linguist. 9 (2021) 53–68.
[21] Y. Tay, D. Bahri, L. Yang, D. Metzler, D. Juan, Sparse sinkhorn attention, in: The 37th International Conference on Machine Learning, Virtual Event, 2020, pp. 9438–9447.
[22] A. Graves, G. Wayne, M. Reynolds, T. Harley, I. Danihelka, A. Grabska-Barwinska, S.G. Colmenarejo, E. Grefenstette, T. Ramalho, J.P. Agapiou, A.P. Badia, K.M. Hermann, Y. Zwols, G. Ostrovski, A. Cain, H. King, C. Summerfield, P. Blunsom, K. Kavukcuoglu, D. Hassabis, Hybrid computing using a neural network with dynamic external memory, Nature 538 (7626) (2016) 471–476.

[23] S. Sukhbaatar, A. Szlam, J. Weston, R. Fergus, End-to-end memory networks, in: Annual Conference on Neural Information Processing Systems, Montreal, Canada, 2015, pp. 2440–2448.
[24] A. Kumar, O. Irsoy, P. Ondruska, M. Iyyer, J. Bradbury, I. Gulrajani, V. Zhong, R. Paulus, R. Socher, Ask me anything: Dynamic memory networks for natural language processing, in: 33nd International Conference on Machine Learning, Vol. 48, New York City, NY, USA, 2016, pp. 1378–1387.
[25] A.H. Miller, A. Fisch, J. Dodge, A. Karimi, A. Bordes, J. Weston, Key-value memory networks for directly reading documents, in: The 2016 Conference on Empirical Methods in Natural Language Processing, Austin, Texas, USA, 2016, pp. 1400–1409.
[26] H. Le, T. Tran, S. Venkatesh, Dual memory neural computer for asynchronous two-view sequential learning, in: The 24th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, London, UK, 2018, pp. 1637–1645.
[27] Y. Wang, W. Chen, D. Pi, L. Yue, S. Wang, M. Xu, Self-supervised adversarial distribution regularization for medication recommendation, in: Thirtieth International Joint Conference on Artificial Intelligence, Virtual Event, 2021, pp. 3134–3140.
[28] J. Song, Y. Wang, S. Tang, Y. Zhang, Z. Chen, Z. Zhang, T. Zhang, F. Wu, Local-global memory neural network for medication prediction, IEEE Trans. Neural Netw. Learn. Syst. 32 (4) (2021) 1723–1736.
[29] S. Lin, M. Wang, C. Shi, Z. Xu, L. Chen, Q. Gao, J. Chen, MR-KPA: medication recommendation by combining knowledge-enhanced pre-training with a deep adversarial network, BMC Bioinform. 23 (1) (2022) 552.
[30] S. Bhoi, M. Lee, W. Hsu, A.H.S. Fang, N.C. Tan, Personalizing medication recommendation with a graph-based approach, ACM Trans. Inf. Syst. 40 (3) (2022) 55:1–55:23.
[31] M. Liu, X. Shen, W. Pan, Deep reinforcement learning for personalized treatment recommendation, Stat. Med. 41 (20) (2022) 4034–4056.
[32] Y. Zhang, R. Chen, J. Tang, W.F. Stewart, J. Sun, LEAP: learning to prescribe effective and safe treatment combinations for multimorbidity, in: 23rd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, Halifax, NS, Canada, 2017, pp. 1315–1324.
[33] L. Wang, W. Zhang, X. He, H. Zha, Supervised reinforcement learning with recurrent neural network for dynamic treatment recommendation, in: 24th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, ACM, London, UK, 2018, pp. 2447–2456.
[34] S. Wang, P. Ren, Z. Chen, Z. Ren, J. Ma, M. de Rijke, Order-free medicine combination prediction with graph convolutional reinforcement learning, in: 28th ACM International Conference on Information and Knowledge Management, ACM, Beijing, China, 2019, pp. 1623–1632.
[35] T.N. Kipf, M. Welling, Semi-supervised classification with graph convolutional networks, in: 5th International Conference on Learning Representations, OpenReview.net, Toulon, France, 2017, pp. 1–10.
[36] A.E. Johnson, T.J. Pollard, L. Shen, H.L. Li-Wei, M. Feng, M. Ghassemi, B. Moody, P. Szolovits, L.A. Celi, R.G. Mark, MIMIC-III, a freely accessible critical care database, Sci. Data 3 (1) (2016) 1–9.
[37] E. Choi, M.T. Bahadori, J. Sun, J. Kulas, A. Schuetz, W.F. Stewart, RETAIN: an interpretable predictive model for healthcare using reverse time attention mechanism, in: Annual Conference on Neural Information Processing Systems, Barcelona, Spain, 2016, pp. 3504–3512.